Craig Fernandes

craig.fernandes@mail.utoronto.ca | 416-436-5210 | https://craigfernandes1.github.io/

Education

PhD Operations Research, University of Toronto — Sep 2021 – May 2026
Advisors: Timothy C.Y. Chan and Ningyuan Chen

MASc Operations Research, University of Toronto — Jan 2020 – Aug 2021
Advisor: Timothy C.Y. Chan

BASc Industrial Engineering, University of Toronto — Sep 2014 – May 2018

Research Visits

Dartmouth College, Tuck School of Business — Summer 2024
Hosts: James Siderius and Raghav Singal

Research (by topic)

Design and Analysis of Innovative Markets

Points Gained in Football: Using Markov Process-Based Value Functions to Assess Team Performance
Timothy Chan, Craig Fernandes, Martin Puterman
Operations Research, 69(3): 877-894, 2021. [DOI, Ungated]
- Media Coverage: INFORMS Analytics Magazine

Income Pools for Superstar Markets
Timothy Chan, Ningyuan Chen, Craig Fernandes
Management Science, 2025 (Articles in Advance). [DOI, SSRN]
- **First Place, CORS Student Paper Competition (Open Category), 2025**
- **First Place, NUS Analytics for X Best Student Presentation, 2025**

Peer Review Market Design: Effort-Based Matching and Admission Control
Craig Fernandes, James Siderius, Raghav Singal
Soon to be Submitted to **Operations Research**, 2025. [SSRN]
- **Accepted, ACM Economics and Computation (EC), 2025**

Handicapping Strategies for Maximizing Engagement in Amateur Darts Tournaments
Timothy Chan, Craig Fernandes, Rachael Walker
Working paper, 2025.
- **First Place, MIT Sloan Sports Analytics Conference Paper Competition, 2024** [Video]
- **Finalist, CORS Student Paper Competition (Undergraduate Category), 2023** [Awarded to Rachael]

Data-Driven B2B Pricing: Learning from Bargainers
Timothy Chan, Ningyuan Chen, Craig Fernandes
Work in progress, 2025.

Estimating Distributions with Failure Rate Properties from Noisy Quantile Data
Timothy Chan, Ningyuan Chen, Craig Fernandes, Muhammad Maaz
Work in progress, 2025.

Analytics Education

Advising Student-Driven Analytics Projects: A Summary of Experiences and Lessons Learned
Aaron Babier, Craig Fernandes, Ian Zhu
INFORMS Transactions on Education, 23(2):121-135, 2022. [DOI]
- Media Coverage: ORMS Today

Moneyball for Murderball: Using Analytics to Construct Lineups in Wheelchair Rugby
Timothy Chan, Craig Fernandes, Albert Loa, Nathan Sandholtz
INFORMS Transactions on Education, 24(2):175-181, 2023. [DOI]
- **First Place, INFORMS Case Competition, 2022**
- Media Coverage: ORMS Today

Sports Analytics

Predicting Plays in the National Football League
Craig Fernandes, Ronen Yakubov, Yuze Li, Amrit Prasad, Timothy Chan
Journal of Sports Analytics, 6(1): 35-43, 2020. [DOI]

A Markov Approach to Untangling Intention Versus Execution in Tennis
Timothy Chan, Doug Fearing, Craig Fernandes, Stephanie Kovalchik
Journal of Quantitative Analysis in Sports, 18(2): 127-145, 2022. [DOI]
- **Finalist, MIT Sloan Sports Analytics Conference Poster Competition, 2022**

Equity, Diversity, and Inclusion in Sports Analytics
Craig Fernandes, Jason Vescovi, Richard Norman, Cheri Bradish, Nathan Taback, Timothy Chan
Journal of Quantitative Analysis in Sports, 20(2): 87-111, 2023. [DOI]

Teaching Experience

Course Instructor

MIE368: Analytics in Action, Average Rating 4.55/5 — Fall 2023

Teaching Assistant

MIE263: Stochastic Operations Research, Average Rating 4.77/5 — Winter 2023, 2024, 2025, 2026
MIE368: Analytics in Action, Average Rating 4.86/5 — Fall 2020, 2021, 2022
- **MIE Teaching Assistant Award, 2022**
- **MIE Group Teaching Assistant Award, 2020**

Invited Seminars

Sloan School of Management (MIT)	Jan 2025
Darden School of Business (Virginia)†	Jan 2025
Zicklin School of Business (Baruch College)†	Dec 2025
HKUST Business School†	Dec 2025
NUS Business School†	Dec 2025
The Wharton School (UPenn)	Dec 2025
IESE Business School	Dec 2025
London Business School	Dec 2025
Imperial Business School	Dec 2025
UCL School of Management	Dec 2025
Mendoza College of Business (Notre Dame)	Nov 2025
Rotman Young Scholar Seminar (Toronto)	May 2025
Tuck School of Business (Dartmouth)	Jul 2024

Conference Presentations

INFORMS Annual Meeting	2022 - 2025
NUS Next-Gen Scholar's Symposium	2025
ACM Economics and Computation (EC)	2025
Marketplace Innovation Workshop (MIW)	2025
INFORMS RMP	2023, 2025
INFORMS MSOM	2023, 2025
CORS Annual Meeting	2018, 2021, 2022, 2025
MIT Sloan Sports Analytics Conference	2022, 2024
New England Symposium on Statistics in Sports	2017, 2021, 2023

Research Mentorship

Undergraduate Thesis Advising (and first position)

Rachael Walker	solo-advised	Analyst, Birch Hill Equity Partners	2023
Mustafa Arif	co-advised with Binghao Zhang	Master's, MIT Sloan	2021
Michael Yin	co-advised with Timothy Chan	Master's, University of British Columbia	2020
Albert Loa	co-advised with Timothy Chan	Master's, University of Toronto	2020

Industry

Amazon, Data Scientist I (Supply Chain Optimization Technologies)	Summer 2021
PepsiCo, Process Improvement Engineering Intern	Summer 2017, 2019

Service

Session Chair

INFORMS Annual Meeting (Revenue Management)	2025
CORS Annual Meeting (Revenue Management)	2025
CORS / INFORMS International (Sports Analytics)	2022

† Cancelled due to logistics.

Scholarships

NSERC Vanier CGS ($150,000)	2023-2025
- **Canada's most competitive PhD scholarship (~4% acceptance rate)**	
NSERC CGS Doctoral [Declined] ($105,000)	2023-2025
TD MDAL Research Grant ($4,000; $3,000)	2023; 2025
Paul Cadario Fellowship in Global Engineering ($8,900)	2025
NSERC CGS Michael Smith Foreign Study Supplement ($6,000)	2023
Mitacs Globalink Research Award ($6,000)	2023
Wallberg Research Fellowship ($7,500)	2022
U of T SGS Conference Travel Grant ($800)	2022
U of T MIE Conference Travel Grant ($650)	2022
Ontario Graduate Scholarship ($15,000)	2021
NSERC CGS Masters ($17,500)	2020
U of T President's Entrance Scholarship ($5,000)	2014

Other Experience

Seminarian, St. Augustine's Seminary — Aug 2015 – June 2019

I lived and studied in a seminary for four years, training to be a Catholic Priest. This involved 30 hours a week of communal and volunteer activities. I overloaded my engineering courses to earn the equivalent of a Minor in Philosophy as preparation for a Master of Divinity, completing one year, before following a different path.

References

Timothy Chan (Co-Supervisor)
Associate Vice-President and Vice-Provost
Professor, Mechanical and Industrial Engineering
University of Toronto
Email: tcy.chan@utoronto.ca

Ningyuan Chen (Co-Supervisor)
Associate Professor, Rotman School of Management
University of Toronto
Email: ningyuan.chen@utoronto.ca

Martin Puterman (Co-Author)
Professor Emeritus, Sauder School of Business
University of British Columbia
Email: martin.puterman@sauder.ubc.ca

Raghav Singal (Co-Author)
Associate Professor, Tuck School of Business
Dartmouth College
Email: raghav.singal@tuck.dartmouth.edu